

Solving Arrays With Sums Using LIA*

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Motivation

- Satisfiability of existential array theories with summation, e.g., for SMT solver
- Past Work: Reduction from arrays with sums and constant arrays to LIA [1]
- Recent approach with sums by Raya [5] reduces to LIA^* , an extension of LIA by a star operator by Piskac and Kuncak [4] from 2008
- But definition of sums in [5] does not align with our definition

Idea

Decide array theories with sums by reduction to LIA^*

Arrays - Intuition

$$a = \begin{array}{|c|c|c|c|} \hline 1 & 8 & -3 & \dots \\ \hline \end{array}$$

- Array of infinite size (unbounded)
- An infinite sequence of entries indicated by $\dots$

Arrays - Intuition

$$a = \begin{array}{|c|c|c|c|} \hline 1 & 8 & -3 & \dots \\ \hline \end{array}$$

$$\sum a = 1 + 8 + (-3) + ???$$

- Array of infinite size (unbounded)
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Arrays - Intuition

$$a = \begin{array}{|c|c|c|c|c|c|} \hline 1 & 8 & -3 & 2 & -5 & 0 \\ \hline \end{array}$$

$$\sum a = 1 + 8 + (-3) + 2 + (-5) + 0$$

- Array of infinite size (unbounded)
- An infinite sequence of entries indicated by $\dots$

LIA*

Definition (LIA)

LIA is the theory of linear integer arithmetic over the signature

$$(\{\mathbb{Z}\}, \{+, -, \dots, -1, 0, 1, \dots\}, \{=, \leq\})$$

For $S \subseteq \mathbb{Z}^n$ define $S^* = \{\sum_{i=1}^n \vec{x}_i \mid n \in \mathbb{N}, \vec{x}_i \in S\}$.

Definition (LIA*)

LIA* is the extension of LIA to formulae of the form

$$F_0(\vec{y}, \vec{z}) \wedge \vec{z} \in \{\vec{x} \mid F(\vec{x})\}^*$$

where F_0, F are quantifier-free LIA formulae, $\vec{x}, \vec{y}, \vec{z}$ vectors of integer variables.

LIA*

Theorem

The satisfiability problem of formulae in LIA is decidable in NP.*

Proof.

Reduction to LIA using a semilinear representation of the star-operator [4]. □

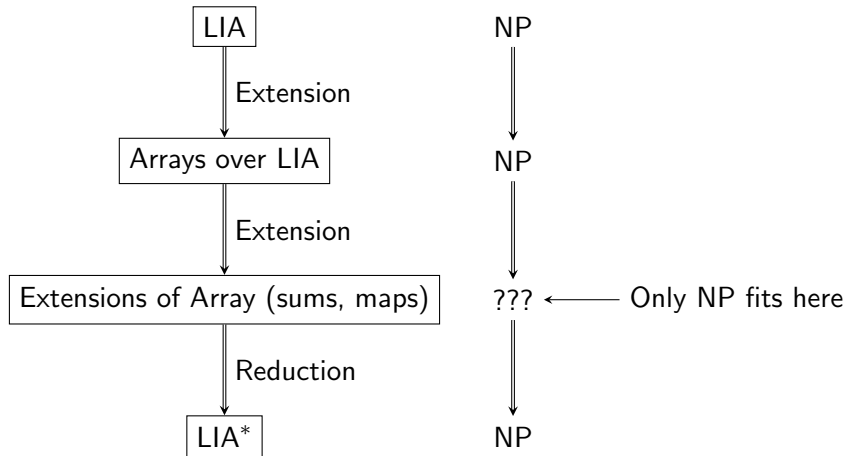
In particular, LIA* does **not** need transitive closure computation.

Example

$$z_2 = z_1 + 3 \wedge \left(\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}\right) \in \left\{\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) \mid x_2 = 2x_1 \vee x_2 = 3x_1\right\}^*$$

is satisfiable for $\left(\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}\right) = \left(\begin{pmatrix} 2 \\ 5 \end{pmatrix}\right)$ since $\left(\begin{pmatrix} 2 \\ 5 \end{pmatrix}\right) = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) + \left(\begin{pmatrix} 1 \\ 3 \end{pmatrix}\right)$ and $2 = 2 \cdot 1, 3 = 3 \cdot 1$.

Roadmap



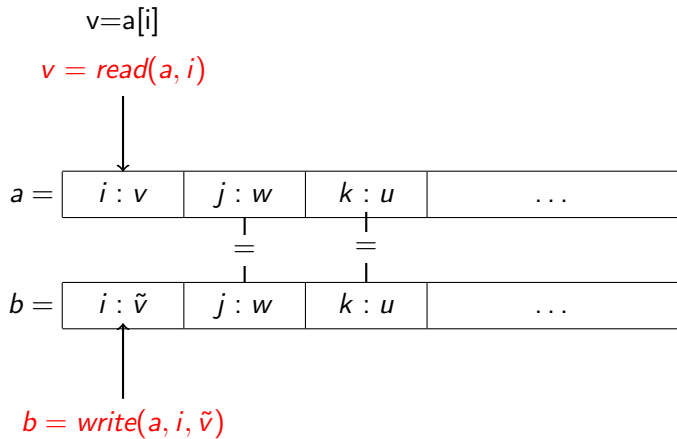
Formalism

- First-order theory of functional arrays defined by *read* and *write* (McCarthy Arrays [2])
- New array sort $\mathbb{Z} \Rightarrow \mathbb{Z}$ and function symbols *read*, *write* in the signature (nested arrays allowed but not treated in this talk)
- Axioms defining *read* and *write*

Formalism

- First-order theory of functional arrays defined by *read* and *write* (McCarthy Arrays [2])
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- Axioms defining *read* and *write*
- There are many extensions (CAL, CaAL, AFL, ...)
- None of the above supports sums

Constraints



Constraints

$$a = K(v) = \begin{array}{|c|c|c|c|} \hline i : v & j : v & k : v & v \\ \hline \end{array}$$

$$b_1 = \begin{array}{|c|c|c|c|} \hline i : v_1 & j : v_2 & k : v_3 & \dots \\ \hline \end{array}$$

$$b = \text{map}_f(b_1, b_2) = \begin{array}{|c|c|c|c|} \hline i : f(v_1, w_1) & j : f(v_2, w_2) & k : f(v_3, w_3) & \dots \\ \hline \end{array}$$

$$b_2 = \begin{array}{|c|c|c|c|} \hline i : w_1 & j : w_2 & k : w_3 & \dots \\ \hline \end{array}$$

Axioms

Read-Over-Write

$$\forall a: (\sigma_1 \Rightarrow \sigma_2), i: \sigma_1, v: \sigma_2.$$

$$write(a, i, v)[i] = v$$

$$\forall a: (\sigma_1 \Rightarrow \sigma_2), i: \sigma_1, j: \sigma_1, v: \sigma_2.$$

$$i = j \vee write(a, i, v)[j] = a[j]$$

Extensionality

$$\forall a: (\sigma_1 \Rightarrow \sigma_2), b: (\sigma_1 \Rightarrow \sigma_2). \exists i: \sigma_1.$$

$$a = b \vee a[i] \neq b[i]$$

Map

$$\forall a_1, \dots, a_n: (\sigma_1 \Rightarrow \sigma_2), i: \sigma_1. map_f(a_1, \dots, a_n)[i] = f(a_1[i], \dots, a_n[i])$$

Constant

$$\forall v: \sigma_2, i: \sigma_1.$$

$$K(v)[i] = v$$

Sum Constraints

Unbounded sum modelled by a **predicate** $sum(\cdot, \cdot)$

$$\text{Intuition : } sum(a, v) \Leftrightarrow \sum_{i \in \mathbb{Z}} a[i] = v$$

If a sum “does not exist” sum evaluates to false.

Sum Axioms

$$\begin{aligned} & \forall i_1, \dots, i_n : I, v_1, \dots, v_n, w : \mathbb{Z}. \text{distinct}(i_1, \dots, i_n) \rightarrow \\ & \quad \left(sum(write(\dots(write(K(0), i_1, v_1) \dots i_n, v_n), w) \Leftrightarrow w = \sum_{i=1}^n v_i \right) \\ & \forall a : (I \Rightarrow \mathbb{Z}), v : \mathbb{Z}. \\ & \quad \left(\exists f : \mathbb{Z} \rightarrow I. (injective(f) \wedge \forall n : \mathbb{Z}. a[f(n)] \neq 0) \right) \rightarrow \neg sum(a, v) \end{aligned}$$

Constraints

- $v = \text{read}(a, i)$
 - $a = \text{write}(b, i, v)$
 - $a = b, a \neq b$
 - $a = K(v)$
 - $a = \text{map}_f(a_1, \dots, a_n)$
 - $\text{sum}(a, v)$
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Theorem

The satisfiability problem of the existential theory of arrays with sums and maps can be reduced to LIA and hence is **decidable in NP**.*

Reduction to LIA*

The hard part is the "..."

$$a = \begin{array}{|c|c|c|c|} \hline 1 & 8 & -3 & \dots \\ \hline \end{array}$$

Example

$$a[i] = 1 \wedge a[j] = 8 \wedge a[k] = -3 \wedge a = \text{map}_f(b) \wedge \text{sum}(a, 3)$$

$$f(x) = \begin{cases} 2x - 2 & x \geq 0 \\ 2x + 3 & x < 0 \end{cases}$$

- Check 1, 8, -3 in the range of f , remains to show $a = \text{map}_f(b) \wedge \text{sum}(a, 3)$
- We need a witness for infinitely many zeroes
- We need values $x_1, \dots, x_n$ in the range of f such that $1 + 8 + (-3) + x_1 + \dots + x_n = 3$, i.e., $x_1 + \dots + x_n = -3$.

Witness for zeroes

$$a = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 1 & 8 & -3 & x_1 & \cdots & x_n & a[i_0] & \cdots \\ \hline \end{array}$$

- Introduce i_0 distinct from all other indices
- Impose $a[i_0] = 0$
- $\Rightarrow$ If it is possible at this position it is possible at infinitely many positions
- Same reasoning as for i, j, k
- Indeed $f(1) = 0$!

$$a = \text{map}_f(b) \wedge \text{sum}(a, 3) \\ \wedge a[i_0] = 0$$

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Values for $x_1, \dots, x_n$

It remains to check that there exists $n \in \mathbb{N}$ and $x_1, \dots, x_n$ such that $x_1 + \dots + x_n = -3$

$$a = \text{map}_f(b) \wedge \text{sum}(a, 3) \wedge a[i_0] = 0$$

$$f(x) = \begin{cases} 2x - 2 & x \geq 0 \\ 2x + 3 & x < 0 \end{cases}$$

$$F(x_a, x_b) \equiv x_a = 2x_b - 2 \wedge x_b \geq 0 \vee x_a = 2x_b + 3 \wedge x_b < 0$$

$$z_a = -3 \wedge \begin{pmatrix} z_a \\ z_b \end{pmatrix} \in \left\{ \begin{pmatrix} x_a \\ x_b \end{pmatrix} \mid F(x_a, x_b) \right\}^*$$

$$\text{Satisfiable for } \begin{pmatrix} z_a \\ z_b \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} -5 \\ -4 \end{pmatrix}$$

The General Case

- Introduce variables z_a, x_a for all arrays a
- Replace $sum(a, v)$ by $a[i_0] = 0$ and $\tilde{v} = z_a$ ($\tilde{v}$ depends on the other reads in the formula)
- Construct F :

$$a = map_f(b_1, \dots, b_n) \rightsquigarrow x_a = f(x_{b_1}, \dots, x_{b_n})$$

$$a = write(b, i, v), a = b \rightsquigarrow x_a = x_b$$

- Add $\vec{z} \in \{\vec{x} \mid F(\vec{x})\}$

Solve resulting formula in LIA*

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Bounded Sums

Definition (Bounded Sum)

$$\text{sum}(a, v, l, u) \leftrightarrow v = \sum_{i=l}^u a[i]$$

Goal: Reduction to $F_0(\vec{y}, \vec{z}) \wedge \vec{z} \in \{\vec{x} \mid F(\vec{x})\}^*$

- Add dummy variables x_d, z_d
- Conjoin $x_d = 1$ to F
- z_d counts the number of summands

$$z_d = \max \{u - l + 1, 0\}$$

Bounded Sums: Example

Example

$$a[3] = l \wedge \text{sum}(a, 6, l, u) \wedge a = \text{map}_f(b)$$

$$f(x) = \begin{cases} 1 & x \geq 4 \\ 0 & x < 4 \end{cases}$$

$$F(x_a, x_b, x_d) \equiv (x_a = 1 \wedge x_b \geq 4 \vee x_a = -1 \wedge x_b < 4) \wedge x_d = 1$$

Case distinction:

- ① $3 \notin \{l, \dots, u\}$: $z_a = 6 \wedge z_d = \max \{0, u - l + 1\} \wedge \vec{z} \in \{\vec{x} \mid F(\vec{x})\}^*$
- ② $3 \in \{l, \dots, u\}$: $z_a = 6 - l \wedge z_d = \max \{0, u - l\} \wedge \vec{z} \in \{\vec{x} \mid F(\vec{x})\}^*$

The first case is not feasible since any solution returns $l < 3 < u$. The second is satisfied for $l = 1, u = 6, a[1] = \dots = a[6] = 1$.

Conclusion and Future Work

Conclusion

- Summation over arrays over the integers can be reduced to LIA^*
- Even if we add map_f -constraints
- Possibly even if we allow bounded sums

Future Work

- Formal proof for any number of bounded sums
- Other theories and fragments with sums
- Generalisation from integers to monoids

- [1] Roland Herrmann and Philipp Rümmer. “What’s Decidable About Arrays With Sums?” In: *Automated Deduction – CADE 30*. Ed. by Clark Barrett and Uwe Waldmann. Cham: Springer Nature Switzerland, 2025, pp. 56–74.
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